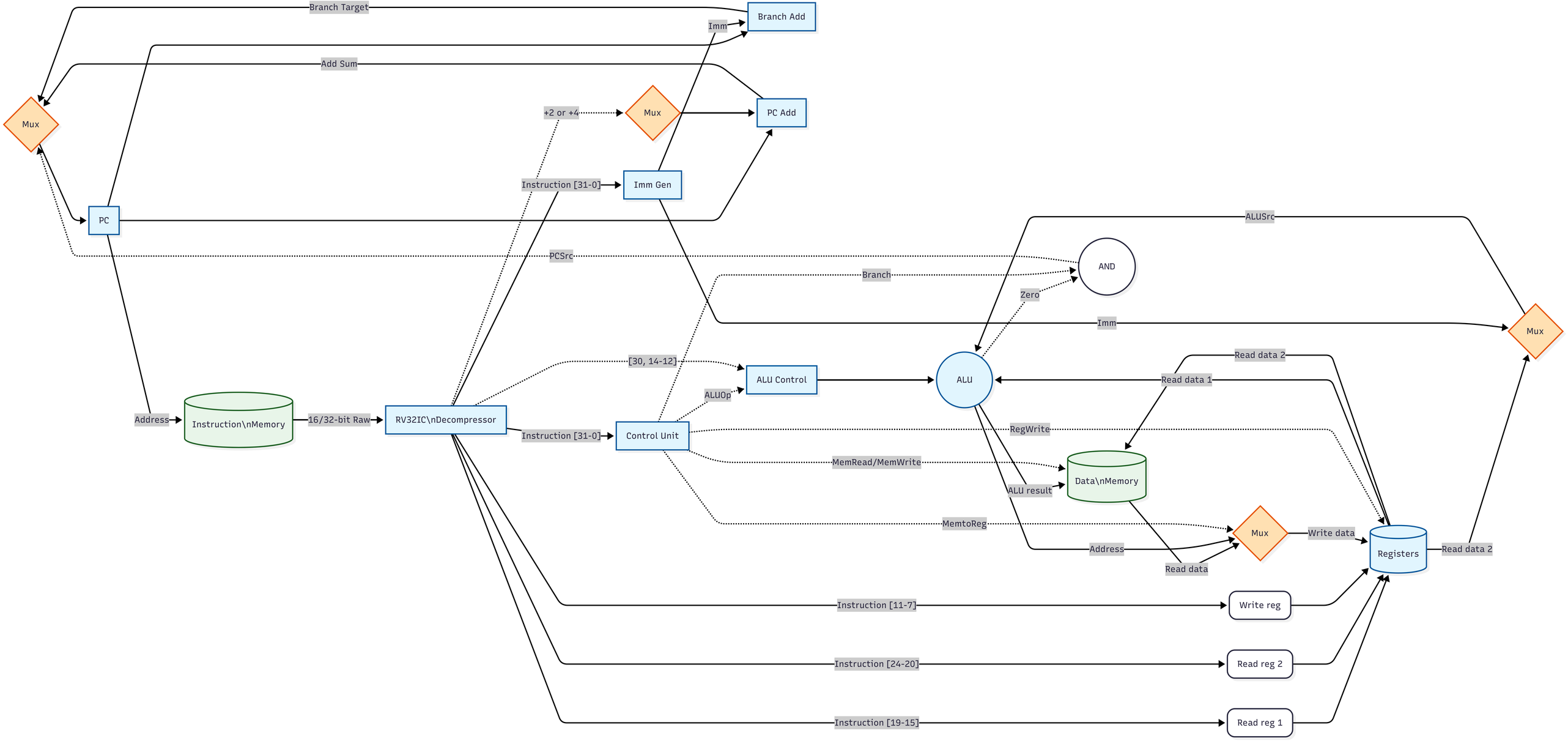
Branch Target
Branch Add
Imm
Add Sum
Mux
+2 or +4
Mux
PC Add
Instruction [31-0]
Imm Gen
PC
ALUSrc
PCSrc
AND
Branch
Zero
Imm
Mux
[30, 14-12]
ALU Control
ALU
Read data 2
Read data 1
ALUOp
Address
Instruction\nMemory
16/32-bit Raw
RV32IC\nDecompressor
Instruction [31-0]
Control Unit
RegWrite
MemRead/MemWrite
Data\nMemory
ALU result
MemtoReg
Mux
Write data
Registers
Read data 2
Address
Read data
Instruction [11-7]
Write reg
Instruction [24-20]
Read reg 2
Instruction [19-15]
Read reg 1